

SΔφ-19

Consciousness and Sleep as Δφ Maintenance Modes

Minimal Axioms for Persistence without Collapse (v1.0)

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This paper opens Cycle-2 of the SΔφ series by formalizing maintenance modes that regulate Δφ persistence. If time is an interpretation of Δφ and closure corresponds to $\Delta\phi \rightarrow 0$, then consciousness can be modeled as a regulator that prevents collapse, while sleep is a maintenance mode that reorganizes traces, reduces drift, and lowers the cost of continued coherence.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Item	Title	DOI
SΔφ-00	Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Minimal Core (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility (v1.1)	10.5281/zenodo.18730448
SΔφ-04	AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min Interpretation (v1.0)	10.5281/zenodo.18735006
SΔφ-08	ARF Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error/Boundary/Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation-Learning-Inference (v1.0)	10.5281/zenodo.18746509

Item	Title	DOI
SΔφ-11	World/Other/Self/Consciousness (v1.0)	10.5281/zenodo.18753403
SΔφ-12	Memory/Forgetting Trace Regulation (v1.0)	10.5281/zenodo.18755685
SΔφ-13	Meaning/Value/Pain Emergence (v1.0)	10.5281/zenodo.18759458
SΔφ-14	Love/Sacrifice/Ethics as Attractor Coupling (v1.0)	10.5281/zenodo.18761928
SΔφ-15	Trust/Hope/Promise as Pre-Commitment Operators (v1.0)	10.5281/zenodo.18762835
SΔφ-16	Evil as Relational Closure (v1.0)	10.5281/zenodo.18763262
SΔφ-17	Openness Stabilization (v1.0)	10.5281/zenodo.18763581
SΔφ-18	Human as the No-Arbiter Species (v1.0)	10.5281/zenodo.18765930

0. Position statement (Cycle-2 entry)

S $\Delta\phi$ -01-18 fixed operational definitions from minimal existence to human as an editable, provisional system under irreversibility. Cycle-2 begins by formalizing maintenance modes that regulate $\Delta\phi$ persistence to prevent collapse into closure. This item models consciousness as a $\Delta\phi$ persistence regulator and sleep as a $\Delta\phi$ maintenance mode.

1. Axiom 19-0 — Time as interpretation of $\Delta\phi$ (recall)

Time is an interpretive output of $\Delta\phi$ via H_T .

$$T = H_T(\Delta\phi).$$

2. Axiom 19-1 — Consciousness as $\Delta\phi$ persistence regulator

Consciousness regulates $\Delta\phi$ so that collapse to $\Delta\phi \rightarrow 0$ is prevented and experience remains available.

$$\text{Consciousness} := \text{Regulate}(\Delta\phi) \text{ such that } \Delta\phi \nrightarrow 0.$$

Minimal functions: (i) mismatch detection, (ii) closure delay via unresolved-state maintenance, (iii) transition-path redistribution.

3. Axiom 19-2 — Sleep as maintenance mode (not collapse)

Sleep is not termination; it is a maintenance mode that reorganizes traces, reduces drift, and prevents closure attractor lock-in.

$$\text{Sleep} := \text{Maintenance}(\Delta\phi) \text{ by reorganizing traces and reducing drift.}$$

4. Axiom 19-3 — Mode switch condition

Switch to sleep when the slope of continued coherence cost exceeds the slope of maintenance/reset cost.

$$\text{Switch} \rightarrow \text{Sleep} \Leftrightarrow d(\text{Cost_coherence})/dt > d(\text{Cost_maintenance})/dt.$$

5. Corollaries (minimal)

C19-A (dream as simulation window, optional): Dream := low-cost rehearsal of transition candidates.

C19-B (link to memory/forgetting): Sleep reorganizes irreversible traces to lower future persistence cost (S $\Delta\phi$ -12).

6. Failure modes (minimal)

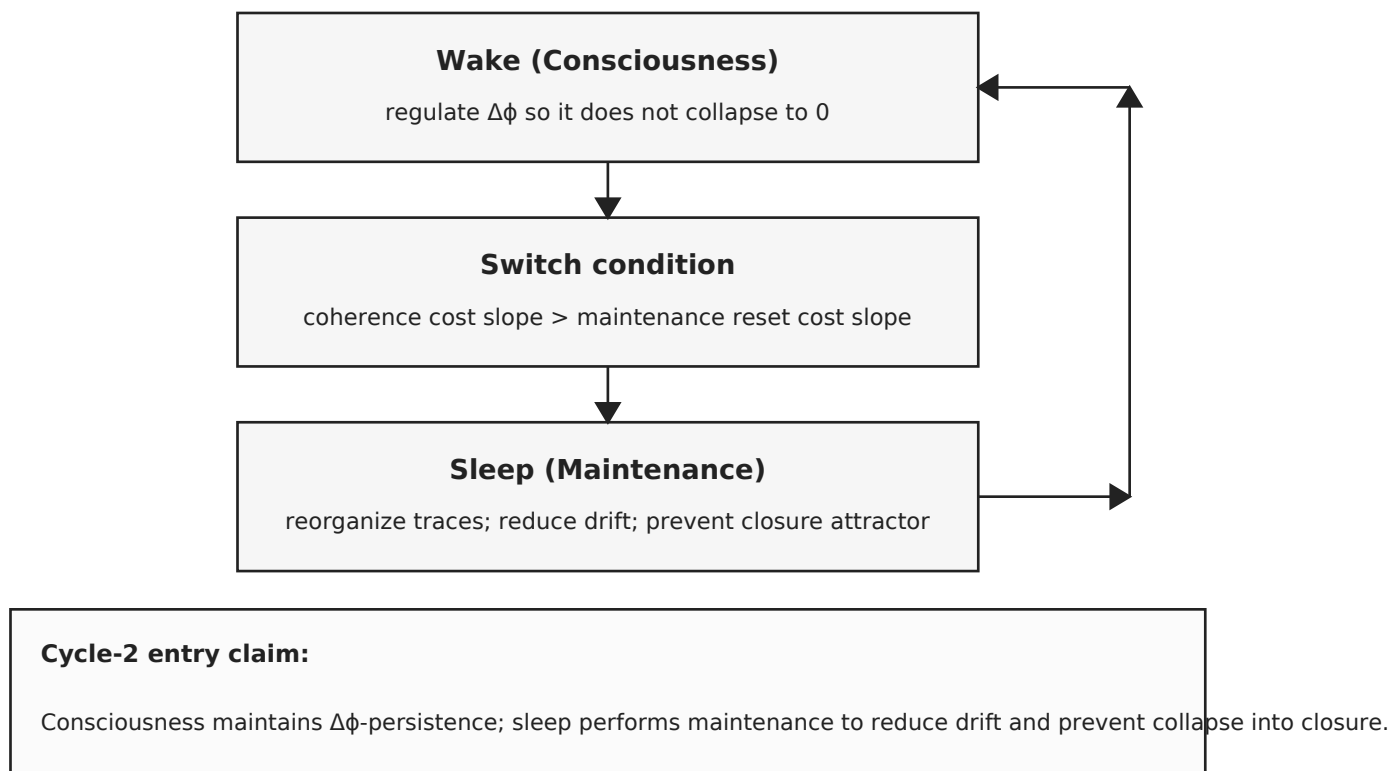
F19-1 (no sleep / over-persistence): coherence cost explodes, increasing closure risk.

F19-2 (excess sleep / under-persistence): $\Delta\phi$ becomes too weak; meaning/value formation fails.

F19-3 (corrupted sleep): maintenance fails; drift increases upon waking.

7. Structural map — $\Delta\phi$ maintenance loop

A minimal loop linking wake persistence regulation and sleep maintenance re-entry.



Note. This document is intentionally non-biological and non-clinical; it fixes only minimal functional maintenance dynamics.

8. Boundary note

This item does not provide neuroscientific mechanisms, clinical guidance, or prescriptions. It fixes only minimal axioms for maintenance modes regulating $\Delta\phi$ persistence.

9. Exit question

If consciousness and sleep are maintenance modes regulating $\Delta\phi$ persistence, what is pain? Is pain the signal of maintenance failure, or the cost of resisting closure?